

Course: Fundamentals of Generative AI and Large Language Models: Theory and Practice

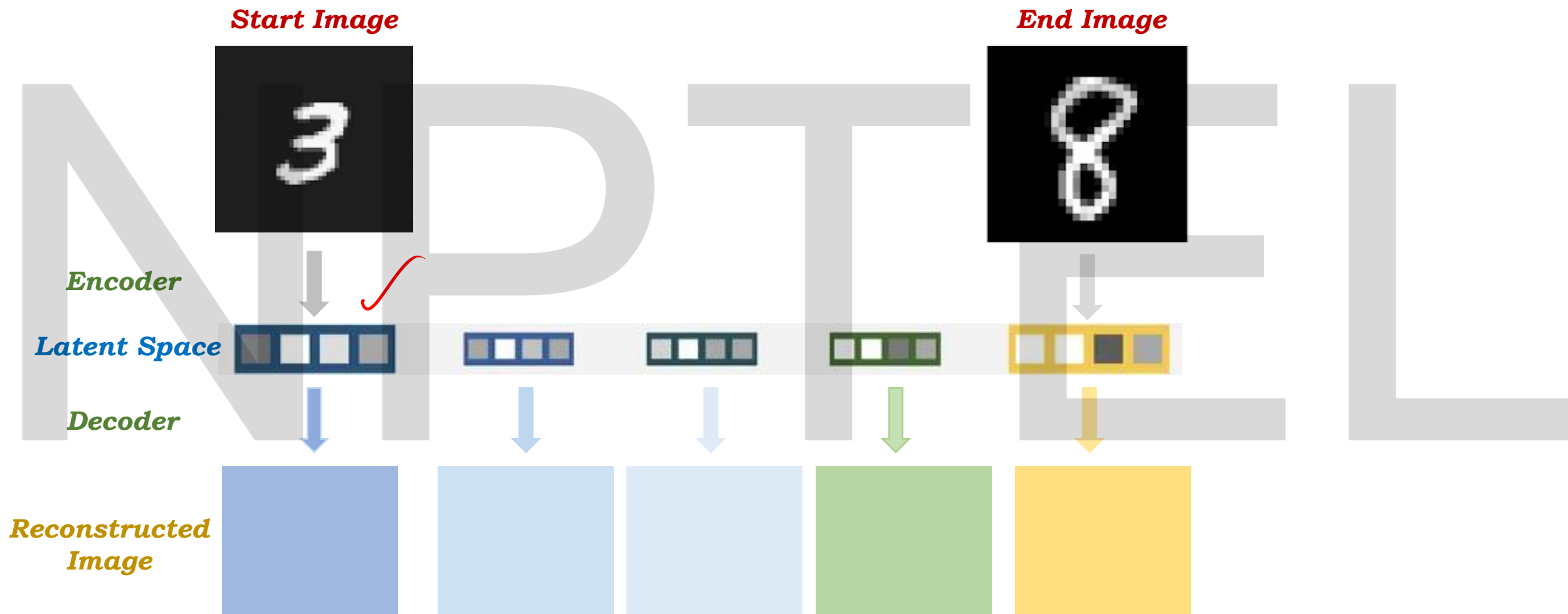
Week 4: Advanced Variational Autoencoders & Practical Applications

Today's Session Overview

➤ Latent space interpolation

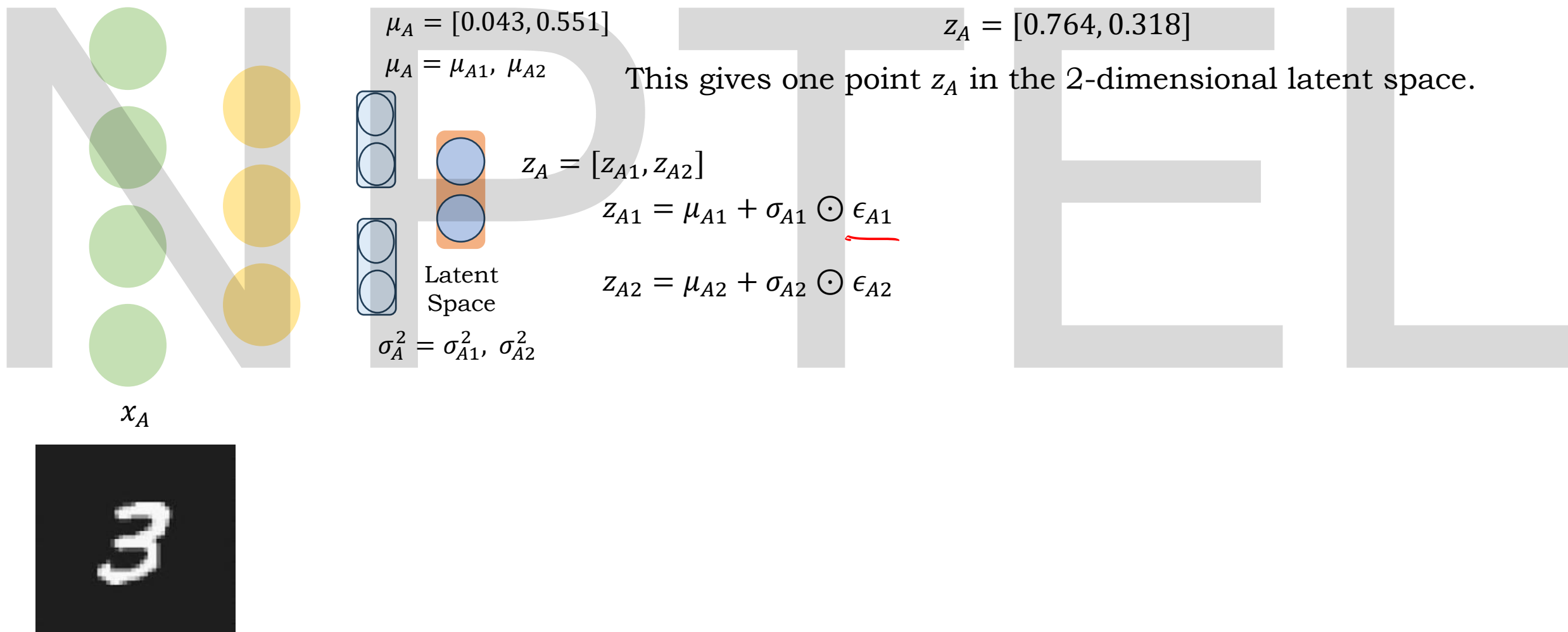
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Latent space interpolation



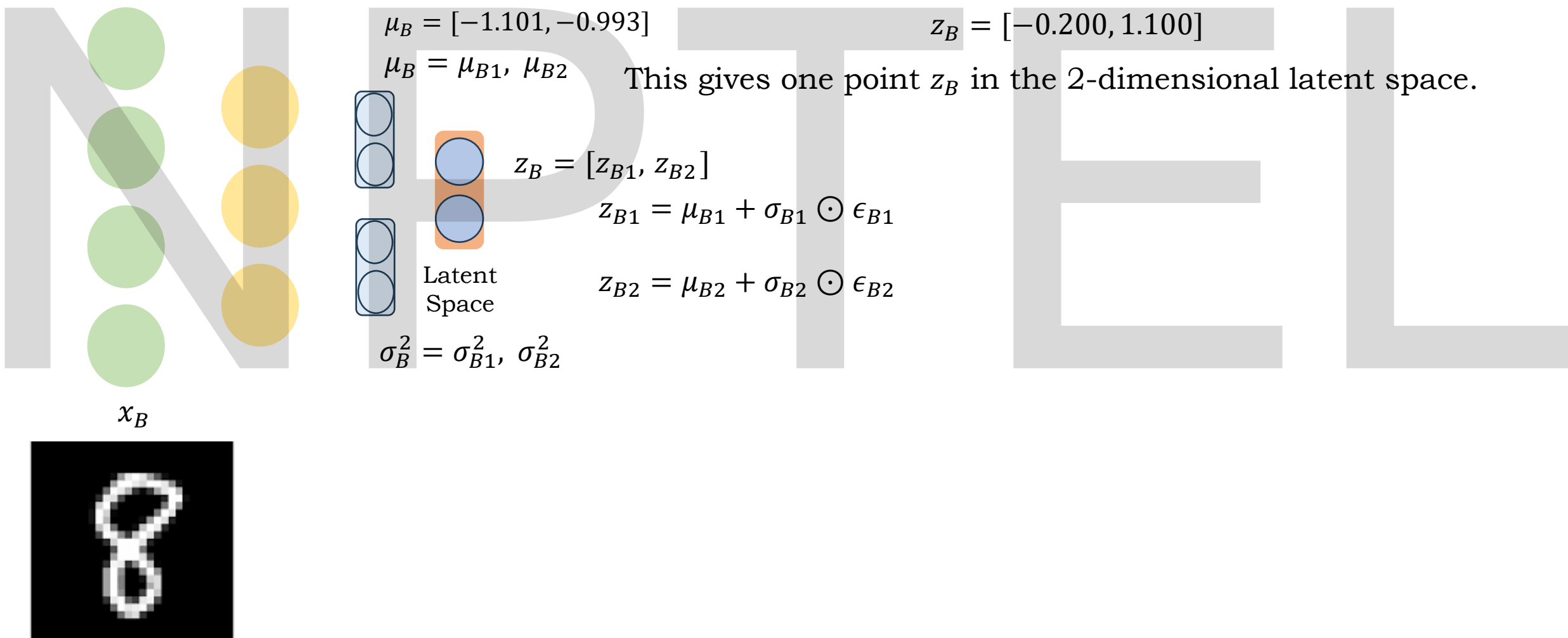
Latent space interpolation

From the input x_A , say we obtained this:



Latent space interpolation

From the input x_B , say we obtained this:



Latent space interpolation

The two latent points:

$$z_A = [0.764, 0.318]$$
$$z_B = [-0.200, 1.100]$$

we perform **latent space interpolation**.

Here, z_A represents image A, digit 3, and z_B represents image B, digit 8.

Step 1: Use interpolation formula

$$z_\alpha = (1 - \alpha)z_A + \alpha z_B$$

where:

α is the interpolation parameter that controls the movement from z_A to z_B .

$$\alpha = 0, 0.25, 0.5, 0.75, 1$$

When $\alpha = 0$, we get exactly z_A , which represents image A.

When $\alpha = 1$, we get exactly z_B , which represents image B.

Intermediate values give points between A and B.

Since latent space is 2-dimensional:

$$z_\alpha = [(1 - \alpha)z_{A1} + \alpha z_{B1}, (1 - \alpha)z_{A2} + \alpha z_{B2}]$$

For $\alpha = 0.25$

$$z_{0.25} = 0.75[0.764, 0.318] + 0.25[-0.200, 1.100]$$

First component: $z_{\alpha,1} = 0.75(0.764) + 0.25(-0.200)$

$$z_{\alpha,1} = 0.573 - 0.050 = 0.523$$

Second component: $z_{\alpha,2} = 0.75(0.318) + 0.25(1.100)$

$$z_{\alpha,2} = 0.2385 + 0.275 = 0.5135$$

So:

$$z_{0.25} = [0.523, 0.514]$$

Latent space interpolation

For $\alpha = 0.5$

$$z_{0.5} = 0.5z_A + 0.5z_B$$

$$z_{0.5} = 0.5[0.764, 0.318] + 0.5[-0.200, 1.100]$$

First component: $z_{\alpha,1} = 0.5(0.764) + 0.5(-0.200)$

$$z_{\alpha,1} = 0.382 - 0.100 = 0.282$$

Second component: $z_{\alpha,2} = 0.5(0.318) + 0.5(1.100)$

$$z_{\alpha,2} = 0.159 + 0.550 = 0.709$$

Therefore:

$$z_{0.5} = [0.282, 0.709]$$

For $\alpha = 0.75$

$$z_{0.75} = 0.25z_A + 0.75z_B$$

$$z_{0.75} = 0.25[0.764, 0.318] + 0.75[-0.200, 1.100]$$

First component: $z_{\alpha,1} = 0.25(0.764) + 0.75(-0.200)$

$$z_{\alpha,1} = 0.191 - 0.150 = 0.041$$

Second component: $z_{\alpha,2} = 0.25(0.318) + 0.75(1.100)$

$$z_{\alpha,2} = 0.0795 + 0.825 = 0.9045$$

Therefore:

$$z_{0.75} = [0.041, 0.9045]$$

What happens after interpolation?

Now each interpolated latent vector is passed to the decoder:

$$\begin{aligned} z_A &\rightarrow \text{Decoder} \rightarrow \hat{x}_A \\ z_{0.25} &\rightarrow \text{Decoder} \rightarrow \hat{x}_{0.25} \\ \underline{z_{0.5}} &\rightarrow \text{Decoder} \rightarrow \underline{\hat{x}_{0.5}} \\ \underline{z_{0.75}} &\rightarrow \text{Decoder} \rightarrow \underline{\hat{x}_{0.75}} \\ \underline{z_B} &\rightarrow \text{Decoder} \rightarrow \underline{\hat{x}_B} \end{aligned}$$

α	Interpolated latent vector z_α
0	[0.764, 0.318]
0.25	[0.523, 0.514]
0.5	[0.282, 0.709]
0.75	[0.041, 0.9045]
1	[-0.200, 1.100]

Conceptually, the generated images will change like this:

Digit 3 → **3 – like intermediate image** → **mixed 3/8 image** → **8 – like intermediate image** → **Digit 8**

This is called **latent space interpolation** because we are moving smoothly between two latent points in the learned latent space.

Latent space interpolation



Gif credit: Julien Despois



Next Session

- Hands-on implementation of VAE

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Thank you